

A Two-Phase Generalized Ewens Law Violates Every Source-Scale Transport–Entropy Bound

Counterexample to the parameter-uniform direct extension in arXiv:1609.07315v3

Status: candidate full counterexample. One explicit strictly positive generalized Ewens law on S_{64} violates each of the source inequalities (7), (8), and (9) with its stated constant. The construction also explains why control uniform over arbitrary cycle weights is impossible.

1. Exact question and the hidden hypothesis

Samson [1] asks whether its preceding transport–entropy inequalities extend to generalized Ewens laws. The tempting observation that these laws are invariant under inversion and conjugation is insufficient: the opening of Theorem 1.3 assumes $\mu \in \mathcal{M}_T$ for the entire theorem, and the proof of part (b) uses the associated product disintegration. The source itself notes that generalized Ewens laws need not lie in $\mathcal{M}$.

$$c(\ell) := \begin{cases} 2 & \text{if } d = d_T. \end{cases}$$

When $\mu = \mu_o$ is the uniform law of a ℓ -local group G , inequalities (7) and (8) hold with

$$c(\ell) := \begin{cases} \ell & \text{if } d = d_H, \\ 1 & \text{if } d = d_T. \end{cases}$$

(b) • Assume that $\mu = \mu_o$ is the uniform law of a ℓ -local group G . Then, for all probability measures ν_1 and ν_2 on G ,

$$(9) \quad \frac{1}{2c(\ell)^2} \widehat{T}_2(\nu_2|\nu_1) \leq \left(\sqrt{H(\nu_1|\mu)} + \sqrt{H(\nu_2|\mu)} \right)^2,$$

with $c(\ell)^2 = 2(\ell - 1)^2 + 2$.

• Assume that G is a normal subgroup of S_n , and that μ satisfies for all $\sigma \in G, t \in S_n$

$$(10) \quad \mu(\sigma) = \mu(\sigma^{-1}) \quad \text{and} \quad \mu(\sigma) = \mu(t^{-1}\sigma t).$$

Then, the inequality (9) holds with $c(\ell)^2 = 8(\ell - 1)^2 + 2$.

The proofs of these results, given in the next section, are inspired by Talagrand seminal work on S_n [Tal95], and Luczak-McDiarmid extension to ℓ -local groups [LM03].

Comments :

- If $G = S_n$ and the class of measure $\mathcal{M}$ is given by (1), the Ewens distribution μ^θ introduced before, is an interesting example of measure in $\mathcal{M}$, satisfying condition (10). This simply follows from its expression given by (2), since for any $\sigma, t \in S_n$, $|\sigma^{-1}| = |\sigma|$ and $|t^{-1}\sigma t| = |\sigma|$.
- An open question is to generalize the above transport-entropy inequalities to the *generalized Ewens distribution* (see the definition in [MNZ12, HNNZ13]). This measure no longer belongs to the class of measure $\mathcal{M}$. In other words, no Chinese restaurant process are known for simulating the generalized Ewens distribution.

2. Idea of the proof: an explicit two-phase law

For generalized Ewens weights $\Theta = (\theta_k)_{k=1}^{64}$, put

$$\theta_1 = 1, \quad \theta_k = 10^{-200} \quad (2 \leq k \leq 63), \quad \theta_{64} = \frac{1}{63!},$$

and let

$$\mu_\Theta(\sigma) = Z^{-1} \prod_{k=1}^{64} \theta_k^{C_k(\sigma)} \quad (\sigma \in S_{64}).$$

Every weight is strictly positive. In unnormalized mass, the identity has weight 1. There are $63!$ many 64-cycles and each has weight $1/63!$, so that conjugacy class also has total mass 1.

Every remaining permutation contains at least one cycle of length in $\{2, \dots, 63\}$ and has raw weight at most $\varepsilon := 10^{-200}$. If r is the aggregate raw mass of all remaining permutations, then

$$0 < r \leq 64! \varepsilon < 10^{-100} < 0.01, \quad Z = 2 + r. \quad (1)$$

Thus μ_Θ is an almost exact half-mixture of the identity and the uniform law on 64-cycles.

3. Simultaneous failure of (7), (8), and (9)

Take $\nu_1 = \delta_e$ and $\nu_2 = \mu_\Theta$. The entropy expression on the right of all two-measure inequalities is

$$E = \left(\sqrt{H(\nu_1 \mid \mu_\Theta)} + \sqrt{H(\nu_2 \mid \mu_\Theta)} \right)^2 = \log(2+r) < \log(2.01) < 0.7. \quad (2)$$

Since the first marginal is a point mass, its coupling with μ_Θ is unique. Every 64-cycle moves all 64 points and has transposition distance 63 from e . Consequently

$$W_{1,H} \geq \frac{64}{2+r}, \quad \tilde{T}_{2,H} = W_{1,H}^2, \quad (3)$$

$$W_{1,T} \geq \frac{63}{2+r}, \quad \tilde{T}_{2,T} = W_{1,T}^2, \quad (4)$$

$$\hat{T}_2(\nu_2 \mid \nu_1) \geq 64 \left(\frac{1}{2+r} \right)^2. \quad (5)$$

The last line follows because a 64-cycle moves each fixed coordinate, with probability at least $1/(2+r)$ under μ_Θ .

For S_{64} one has $\ell = 2$ and $K_n = 63$. In part (a), $c = 3$ for Hamming distance and $c = 2$ for transposition distance. The right side of (7) and (8) is below $63(0.7) = 44.1$. Using only $r < 0.01$, their left sides satisfy

source bound	cost	left side lower bound
(7)	$W_{1,H}$	$\frac{2}{9}(64/2.01)^2 > 225$
(7)	$W_{1,T}$	$\frac{1}{2}(63/2.01)^2 > 491$
(8)	$\tilde{T}_{2,H}$	$\frac{1}{18}(64/2.01)^2 > 56$
(8)	$\tilde{T}_{2,T}$	$\frac{1}{8}(63/2.01)^2 > 122.$

Hence all four versions fail.

For the invariant-law constant in (9), $c(2)^2 = 8(2-1)^2 + 2 = 10$. Equations (2) and (5) give

$$\frac{1}{2c(2)^2} \hat{T}_2(\nu_2 \mid \nu_1) > \frac{64}{20(2.01)^2} > 0.79 > 0.7 > E,$$

so (9) fails as well. This proves the promised simultaneous counterexample.

4. Scope and stronger obstruction

The counterexample refutes the direct extension with constants uniform over all positive cycle weights. Replacing 64 by growing n makes the squared metric cost order n^2 while the entropy stays near $\log 2$; hence the same family defeats every parameter-independent $O(n)$ constant.

It does not preclude transport inequalities with constants depending on Θ , nor results under quantitative regularity assumptions on (θ_k) . Such dependence is unavoidable: on a finite group every positive law admits some finite constant, while arbitrary cycle weights can create the separated phases above.

References

- [1] P.-M. Samson, *Transport-entropy inequalities on locally acting groups of permutations*, arXiv:1609.07315v3, pp. 5–6.